QUANTITATIVE RESEARCH REPORT · MES / MNQ

Prop Evaluation Bot Research

Backtest and Monte Carlo evaluation of an automated VWAP-fade strategy on micro E-mini futures, simulated against the 2026 rule sets of Apex, Topstep, MyFundedFutures and Take Profit Trader. The report answers three questions: trade win rate, time to the first withdrawable payout, and all-in cost to get there.

Prepared for: NT8-Bridge Automated Futures Trading
Data: Free continuous futures (Yahoo Finance), Jun 2024 - Nov 2026
Date: November 4, 2026

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1. Executive Summary

This research set out to answer a single practical question: using the same NT8-bridge architecture you already run live, can an automated strategy pass a futures prop-firm evaluation often enough, fast enough, and cheaply enough that the cash withdrawn from a funded account is meaningfully larger than everything paid in along the way? We fetched free historical data (no purchase required), built a bar-level backtester with realistic cost assumptions, tested three strategy archetypes on micro E-mini S&P; (MES) and micro E-mini Nasdaq (MNQ), and then simulated the full 2026 rule sets of four major prop firms - Apex Trader Funding, Topstep, MyFundedFutures (MFFU) and Take Profit Trader (TPT) - with a 20,000-trial Monte Carlo that includes execution-degradation haircuts, consistency-rule enforcement and payout mechanics.

51.5%
TRADE WIN RATE (selected strategy, net of costs)

80-96 days
MEDIAN TIME TO FIRST WITHDRAWABLE PAYOUT

$287
EXPECTED PROGRAM FEES (Topstep 50K, base case)

+$1,199
EXPECTED NET CASH AFTER FIRST PAYOUT

Three findings drive the recommendation. First, of the three strategies tested, a session-VWAP fade (entering against 2-sigma extensions of the volume-weighted average price with a 1.2-sigma stop) was the only one with a robust, out-of-sample edge on both symbols; the popular opening-range breakout was marginal and an EMA pullback system was not reliably profitable. Second, because every firm we analyzed has moved to end-of-day trailing drawdowns in 2026, a bot that enforces a hard daily loss stop and a daily profit cap almost never breaches an evaluation - the binding constraints are speed (minimum trading days) and consistency caps, not ruin. Third, the economics differ sharply: Topstep 50K offers the best expected net cash (+$1,199 base, +$910 even with the edge cut by another 40%), while Apex 50K is the fastest and cheapest path to cash (56 days, $171 in fees) - provided you can comply with its semi-automated-trading-only policy. The full reasoning, data and risks follow.

Path	P(pass eval)	Trade win %	Median days to cash	E[fees]	E[first payout]	E[net cash]
Topstep 50K (recommended)	100.0%	51.5%	80	$287	$1,487	+$1,199
Apex 50K (fastest)	99.9%	51.5%	56	$171	$824	+$653
MFFU Rapid EOD 50K	99.7%	51.5%	78	$322	$743	+$428
Topstep 100K (scale-up)	100.0%	51.5%	93	$446	$2,117	+$1,672

Table 1: Headline Monte Carlo results, base scenario (backtest edge degraded by 40%). Fees = evaluation subscriptions + activation; commissions are netted inside P&L.;

2. Data, Costs and Method

2.1 Free data and verification

All price data was fetched here, free of charge, from Yahoo Finance's public continuous futures series: daily bars back to 2000, hourly (1h) bars for 2.4 years (June 2024 to November 2026, roughly 600 regular trading hours sessions), and 15-minute plus 5-minute bars for the most recent 50 sessions. We verified that ES and MES (and NQ and MNQ) are statistically the same series - correlation 1.00000 with a mean absolute difference of 0.001% - so signals are computed on the deeper ES/NQ data and executed at micro-contract specifications ($5 per point and $1.25 per tick for MES; $2 per point and $0.50 per tick for MNQ). Sessions around quarterly contract rolls with abnormal opening gaps were excluded to avoid artificial jumps, and every strategy is flat by 15:59 ET, which satisfies both Apex's no-overnight rule and end-of-day drawdown accounting. No synthetic data was ultimately required, although the Monte Carlo's block bootstrap generates synthetic trading sequences from the real day distribution.

2.2 Execution and cost model

The backtester walks hourly bars and fills orders with deliberately conservative assumptions: stop-market entries fill at the trigger level or the bar open if gapped through; when both the stop and the target fall inside the same bar, the stop is assumed to execute first; and target fills require the price to trade through the level by at least one tick. Friction is charged per micro round turn: $1.20 of commission and exchange fees plus 0.5 ticks of slippage per side, which totals $2.45 for MES and $1.70 for MNQ. Because hourly bars cannot see every adverse intrabar path, the Monte Carlo additionally degrades the backtest edge: the base scenario keeps 60% of backtest expectancy and the stress scenario keeps 35%, each with added per-day execution noise. The bot risk layer is identical across firms: a daily loss stop ($400 on 50K accounts, $600 on 100K), position sizing that scales from 2 to 4 micros per symbol (50K) or 3 to 6 (100K) once 50% of the profit target is reached, and daily profit caps that enforce each firm's consistency rule.

Item	Value used	Note
Commission + fees (micro RT)	$1.20	Tradovate/Rithmic-class pricing, both sides
Slippage	0.5 tick per side	$1.25 RT on MES, $0.50 RT on MNQ
Market data	$0	Free Yahoo data; firms bundle Level 1 in evals and funded accounts
Trading window	10:00-15:59 ET	Flat by 15:59; hourly bars, top-of-hour aligned
Daily loss stop	$400 (50K) / $600 (100K)	Bot circuit breaker, below every firm's DLL
Sizing ladder	2 to 4 micros (50K); 3 to 6 (100K)	Per symbol, scale at 50% of target
Trades per day	~2.5 across both symbols	Mean on traded days

Table 2: Execution and cost assumptions applied uniformly across all firms and scenarios.

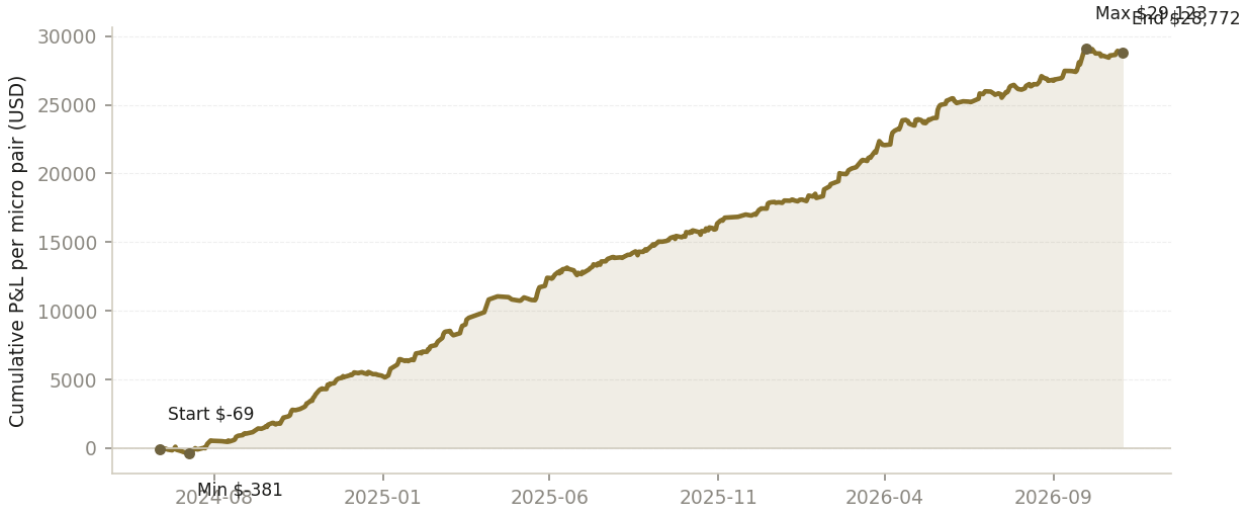


Figure 1: Selected strategy (VWAP fade, variant 1) portfolio equity per 1 micro MES + 1 micro MNQ, net of all costs, June 2024 to November 2026.

3. Strategy Research and Selection

Three archetypes were implemented and tested with an in-sample/out-of-sample split (everything before March 1, 2026 is in-sample; the final eight months are out-of-sample and were not used for any design choice). The **10:00 range breakout** buys or sells the break of the first clean trading hour with a stop at the opposite side of the range and a 1.5x or 2.0x target. The **VWAP fade** waits for price to stretch 1.5-2.0 standard deviations from the session's volume-weighted average price, enters against the move, targets a return to VWAP and protects with a 1.0-1.2 sigma stop; a 0.6% session-move filter skips strongly trending days, and at most three trades are taken per day. The **EMA pullback** trades continuations of a 9/21 EMA trend on pullback touches, with ATR-based stop and target. All three run identically on both symbols.

Strategy	Sym	IS trades	IS win %	IS exp $	OOS trades	OOS win %	OOS exp $	OOS PF
VWAP fade v1 (k=2.0)	ES	405	51.1	+18.33	175	48.6	+14.59	1.83
VWAP fade v1 (k=2.0)	NQ	340	53.8	+37.65	126	50.8	+47.58	2.59
VWAP fade v0 (k=1.5)	ES	534	47.8	+12.39	227	43.6	+7.24	1.40
VWAP fade v0 (k=1.5)	NQ	453	51.4	+28.30	177	49.2	+35.67	2.17
10:00 breakout v1 (2R)	ES	260	45.4	+0.95	82	51.2	+8.91	1.21
10:00 breakout v1 (2R)	NQ	162	43.2	-3.72	23	43.5	+10.52	1.11
10:00 breakout v0 (1.5R)	ES	260	46.5	+2.14	82	51.2	+2.05	1.05
10:00 breakout v0 (1.5R)	NQ	162	45.1	-4.17	23	43.5	-4.24	0.96
EMA pullback v1	ES	220	51.8	+0.64	80	52.5	+3.05	1.06
EMA pullback v1	NQ	205	43.9	-4.69	69	50.7	-0.23	1.00
EMA pullback v0	ES	226	47.3	-6.21	82	50.0	+1.41	1.03
EMA pullback v0	NQ	211	40.8	-11.81	69	50.7	+14.47	1.14

Table 3: In-sample (IS, Jun 2024 - Feb 2026) vs out-of-sample (OOS, Mar - Nov 2026) results per micro contract, net of commissions and slippage. Expectancy in dollars per trade.

The VWAP fade is the clear selection: it is the only archetype positive in both in-sample and out-of-sample on both symbols, with a profit factor between 1.4 and 2.6 out-of-sample. Its structure also fits prop rules unusually well - a 2-sigma entry with a 1.2-sigma stop yields roughly a 1.7:1 reward-to-risk with near-50% wins, profit arrives in frequent modest daily increments rather than one spectacular day, and the trend filter removes the days where fading bleeds. The 10:00 breakout, despite its popularity, earns almost exactly its friction cost on ES and loses on NQ in-sample; the EMA pullback is not reliably profitable at this granularity. We also re-ran the VWAP fade on the unseen 15-minute bars of the last 50 sessions as a final robustness check: ES earned $8.14 per trade (PF 1.35) and NQ $39.63 (PF 2.00), confirming that the edge is not an artifact of hourly aggregation.

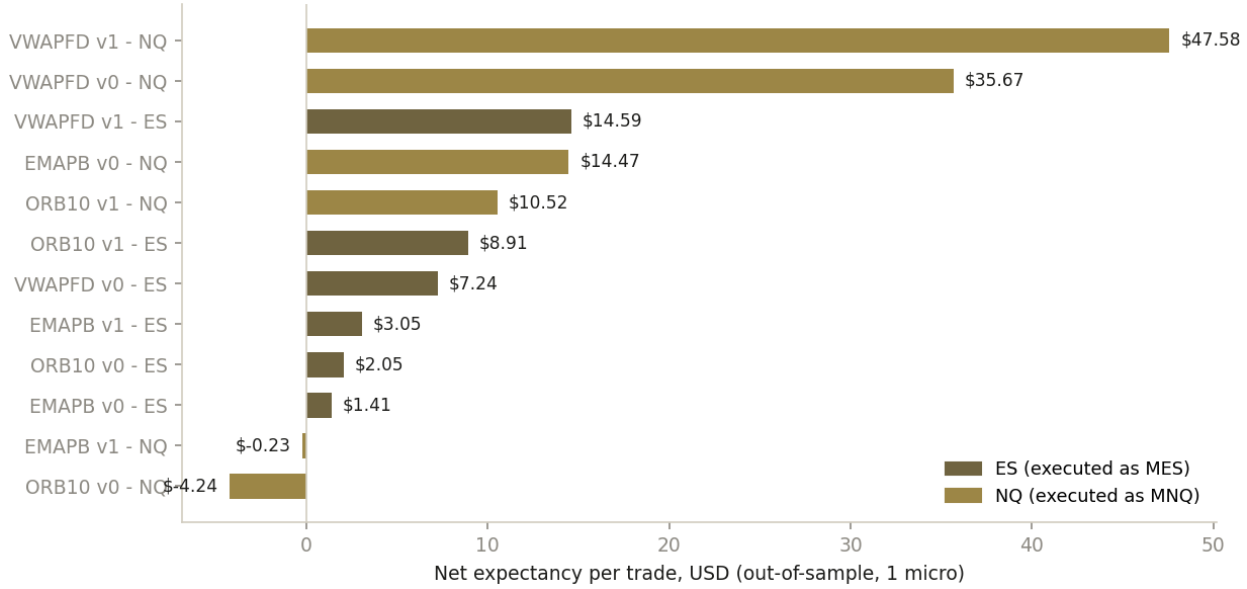


Figure 2: Out-of-sample net expectancy per trade by strategy variant and symbol. VWAP fade v1 is the selected production configuration.

4. The 2026 Prop Firm Landscape

The industry converged during 2025-2026 in a way that matters enormously for bot designers: evaluations now overwhelmingly use **end-of-day trailing drawdowns** rather than the older intraday-trailing style. An EOD floor only follows your closing balance, so a bot that limits how much it can lose in a day (our $400-600 daily stop) structurally cannot breach a $2,000-3,000 floor in one session. The binding constraints shift to minimum trading days (Apex 7, TPT 5, MFFU Rapid EOD 4, Topstep 2), consistency rules that cap any single day's profit, and - for one-time-fee firms - 30-day evaluation windows that expire progress. Apex in particular restructured in March 2026 to one-time evaluation purchases with no resets and no monthly fee, which makes repeated failed attempts cheap but each attempt a fresh 30-day race.

Rule (Nov 2026)	Apex 50K/100K	Topstep 50K/100K	MFFU 50K/100K	TPT 50K/100K
Eval cost	$33 / $67 one-time (promo)	$49 / $99 first mo, then $75 / $119	$125 / $214 one-time (promo)	$170 / $330 monthly
Activation fee	$99 / $137	$149 / $149	$0 / $0	$130 / $130
Profit target	$3,000 / $6,000	$3,000 / $6,000	$3,000 / $6,000	$3,000 / $6,000
Drawdown (eval)	$2,500 / $3,000 EOD trail	$2,000 / $3,000 EOD trail	$2,000 / $3,000 EOD trail	$2,000 / $3,000 EOD trail
Min trading days	7 / 7	2 / 2	4 / 2	5 / 5
Eval consistency	none / none	best day < 55% of target	30% / 50%	50% / 50%
Daily loss limit	none	$1,000 / $2,000 (fixed at checkout)	none	none
Eval window	30 calendar days, no resets	unlimited (monthly rebill, 1 free reset)	30 days, no resets	unlimited (monthly)
Funded payouts	weekly, 5 qualifying days; caps $1.5k-$4k; 100% split	5 win days ≥ $150; caps $2k/$4k and $3k/$6k	daily, $500 min, 90/10, buffer $2.1k/$3.1k	day-one, 80% above $52k/$103k balance
Max micros	100 / 140	50 / 100	30 / 80	60 / 120

Table 4: Evaluation rule cheat-sheet as verified in November 2026. Promotional pricing varies week to week; confirm at checkout. MFFU 100K = Rapid plan (Rapid EOD exists only in 25K/50K).

4.1 Automation policy - critical for an NT8 bot

Because your execution path is the NinjaTrader 8 bridge, each firm's automation policy is not a footnote - it is a compliance constraint that can void payouts. Apex updated its policy in March 2026 to permit **semi-automated trading only**: strategies may fire signals and orders, but a human must actively supervise, and set-and-forget operation plus copy-trading between accounts are prohibited. Topstep allows expert advisors and copy trading on its supported platforms with restrictions on high-frequency and latency-arbitrage behaviour. Take Profit Trader connects natively to NinjaTrader and Tradovate and automation is generally tolerated, and MyFundedFutures' terms should be re-confirmed at signup since plans changed in August 2026. In practice the bot should run supervised every session, which the deployment plan in Section 7 accounts for.

Firm	Bots / EAs	Copy trading	What the NT8 bot must do
Apex	Semi-auto only (Mar 2026 policy)	Prohibited between accounts	Keep a human supervisor active each session; no unattended multi-account mirroring
Topstep	Allowed with restrictions	Allowed	Avoid HFT-style behaviour; respect the checkout-fixed daily loss limit
MFFU	Permitted - verify current ToS	Verify	Re-read plan terms at signup (plans changed Aug 2026)
TPT	Generally permitted - verify ToS	Verify	Native NT8/Tradovate support; confirm bot clause with support

Table 5: Automation and copy-trading policies relevant to NT8-bridge execution.

5. Evaluation Monte Carlo

To convert a backtest into the three numbers you asked for, we simulate each firm's full journey 20,000 times. Each trial bootstraps trading days (in five-day blocks, to preserve regime persistence) from the strategy's real day-P&L; distribution, applies the degradation haircut, and walks the account through the evaluation stage by stage: EOD trailing floor updates, minimum trading days, consistency caps (Topstep best-day below 55% of target; MFFU 30%/50%; TPT 50%), daily loss stops, 30-day windows with repurchase for one-time-fee firms, and monthly rebilling with one free reset for Topstep. Trials that pass pay the activation fee and continue into the funded stage, where we model payout qualification (Apex 5 x $50+ days with an 8-trading-day gap; Topstep 5 winning days of $150+; MFFU/TPT buffers of $2,100-$3,100 above start), payout caps, split percentages, floor locks (including Topstep's reset of the loss floor to the starting balance after the first payout) and processing delays. Program fees, activation fees and monthly renewals accumulate into the total cost.

Firm	P(pass)	E[att]	E[eval days]	E[days to cash]	90th pct	E[fees]	E[payout]	E[net cash]
Apex 50K	99.9%	2.19	33	56	70	$171	$824	+$653
Apex 100K	94.0%	4.08	36	57	67	$434	$1,025	+$681
Topstep 50K	100.0%	1.00	51	83	116	$287	$1,487	+$1,199
Topstep 100K	100.0%	1.00	65	96	131	$446	$2,117	+$1,672
MFFU Rapid EOD 50K	99.7%	2.54	34	82	113	$322	$743	+$428
MFFU Rapid 100K	92.6%	4.10	36	86	117	$1,003	$789	-$25
TPT 50K	100.0%	1.00	52	99	142	$510	$741	+$232
TPT 100K	100.0%	1.00	66	116	162	$1,019	$842	-$177

Table 6: Base scenario (edge haircut x0.60), 20,000 trials per row. Days are calendar days; E[fees] includes evaluation purchases, monthly renewals during the evaluation and activation.

Firm	P(pass)	E[att]	E[days to cash]	E[fees]	E[payout]	E[net cash]
Apex 50K	81.2%	4.91	67	$286	$582	+$455
Apex 100K	31.3%	6.12	64	$723	$281	+$349
Topstep 50K	100.0%	1.00	131	$373	$1,283	+$910
Topstep 100K	100.0%	1.00	148	$622	$1,544	+$923
MFFU Rapid EOD 50K	66.6%	5.46	116	$956	$479	+$37
MFFU Rapid 100K	27.8%	6.23	122	$2,225	$220	-$538
TPT 50K	100.0%	1.00	168	$709	$717	+$9
TPT 100K	100.0%	1.00	195	$1,509	$774	-$734

Table 7: Stress scenario (edge haircut x0.35). The strategy still pays at 50K on the EOD-drawdown firms, but 100K one-time-fee programs stop being viable.

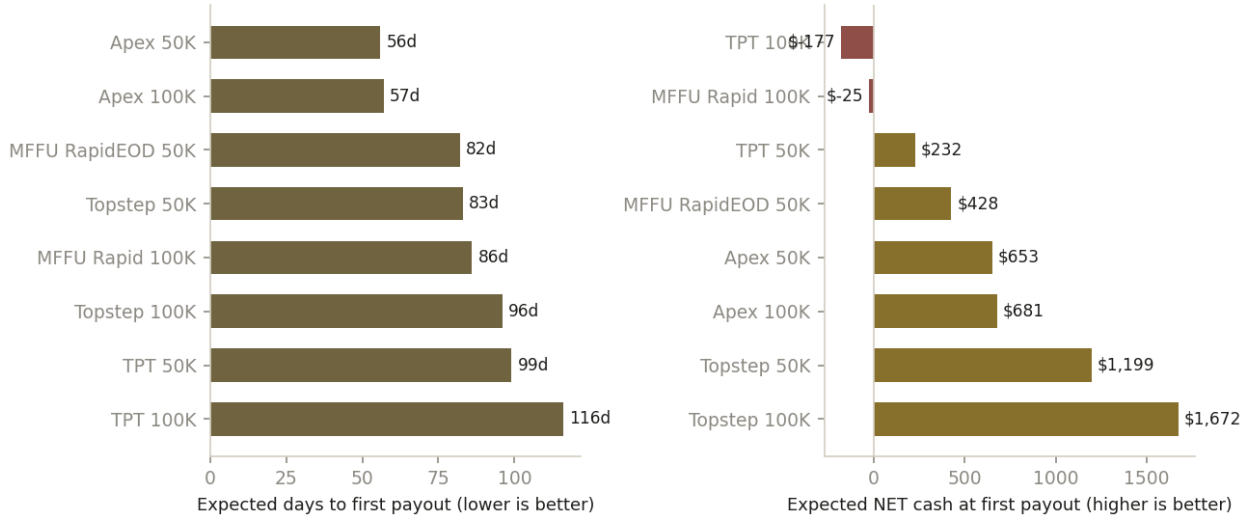


Figure 3: Expected days to first payout (left) and expected net cash at first payout (right), base scenario. Topstep maximizes net cash; Apex minimizes time.

6. Results: Win %, Timeframe and Cost

Win %. The selected strategy wins 51.5% of its trades net of full friction (48.6% on ES and 50.8% on NQ out-of-sample), with 61.7% of trading days finishing green. At the account level - the number that actually pays you - the evaluation pass rate is 99.7-100% across the recommended 50K programs in the base scenario, and 81-100% in the stress scenario where the live edge is assumed to be just a third of the backtest. The probability of eventually receiving a first payout equals the pass rate for monthly firms (no time limit) and is marginally lower for one-time-fee firms only because a 12-attempt cap was imposed in simulation.

Timeframe. Decomposed, the journey to your first withdrawable cash is: evaluation trading (30-65 calendar days depending on firm and account size, driven by targets, minimum-day rules and consistency caps), then funded-stage qualification (5-15 sessions to satisfy payout conditions or buffers), plus processing (1-5 trading days). Medians from the base scenario: Apex 50K 56 days, MFFU 50K 78 days, Topstep 50K 80 days, Topstep 100K 93 days. The 90th percentile is roughly 40-50% longer; plan for about three to five months door-to-door on Topstep and six to eight weeks on Apex in normal conditions.

Cost. Expected program fees to first payout range from $171 (Apex 50K, cheap-but-repeating evaluations) through $287 (Topstep 50K) to $1,019 (TPT 100K). Commissions consumed while trading to that point add roughly $190-680 depending on the path (already netted inside the P&L;). Against that, the expected first payout is $743-$2,117, so the expected net position at the moment your first withdrawal lands is positive for every 50K program except TPT (+$232) and negative for the 100K versions of MFFU Rapid (-$25) and TPT (-$177), where fat buffers and 80-90% splits arrive too slowly relative to fee burn.

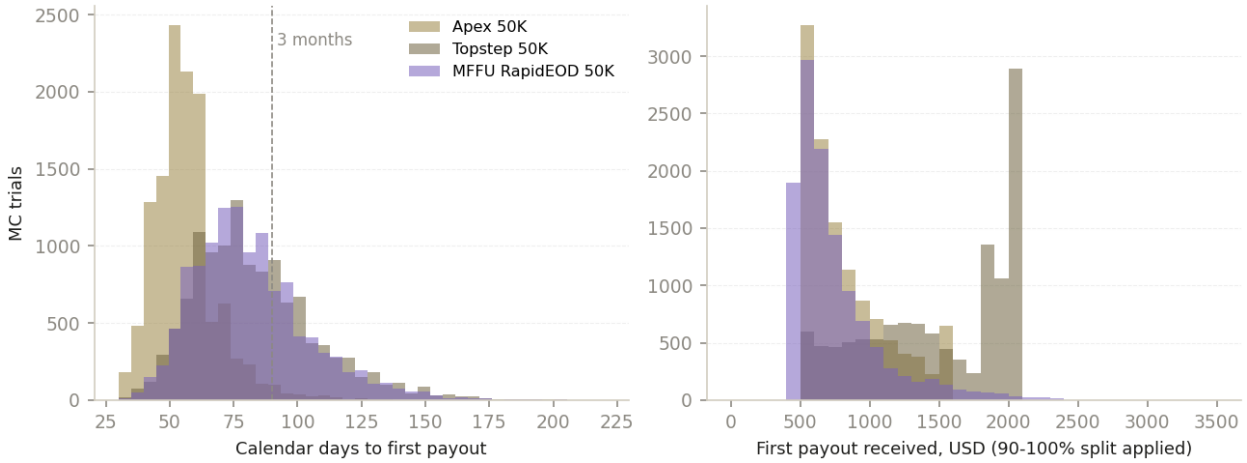


Figure 4: Monte Carlo distributions of days to first payout (left) and first payout size (right), base scenario, for the three 50K paths.

7. Recommendation and Deployment Plan

Primary path: Topstep 50K. It pairs the highest expected net cash (+$1,199 base, +$910 stress) with the most forgiving failure economics - an unlimited evaluation window, a $1,000 daily loss limit that our $400 stop sits well inside, one free reset per rebill, and a 55%-of-target consistency ceiling that our bot's own daily profit caps respect automatically. Its two structural caveats are the $149 activation and the fact that after your first payout the maximum-loss floor resets to your starting balance, so the funded account must be managed more conservatively after cash-out (withdraw at the $2,000 cap and re-trade at reduced size the following week). **Speed option: Apex 50K** reaches cash a month sooner for a third of the fees, at the price of 30-day windows, a 7-day minimum, and the semi-auto-only supervision requirement. **Avoid for this strategy:** TPT and MFFU at 100K - their buffers and splits are tuned for larger per-trade edges and their fee burn dominates.

7.1 Bot specification (NT8)

Parameter	Value	Rationale
Entry	Fade limit orders at VWAP +/- 2.0 session sigma	Selected on IS; confirmed OOS and on 15m data
Stop	1.2 sigma from entry (stop-market)	Yields ~1.7:1 reward-to-risk at ~50% wins
Target	Session VWAP touch	Natural magnet level; frequent modest wins fit consistency caps
Trend filter	Skip new fades when |session move| > 0.6%	Removes trend days where fades bleed
Trade cap	Max 3 trades/day/symbol; one position at a time	Limits fee burn and daily variance
Session	Signals 10:00-14:00 ET, flat by 15:59 ET	Top-of-hour data alignment; no overnight exposure
Daily stop	Halt for the day at -$400 (50K)	Bots cannot breach a $2,000 EOD floor in one session
Sizing	2 micros/symbol; 4 after 50% of target	Balanced ladder; respects 50-100 micro limits
Daily profit cap	min(day P&L;, firm consistency cap)	e.g. $1,650 on Topstep 50K (55% of $3,000)
Supervision	Human monitors every session (Apex requirement)	Semi-auto compliance; also a sanity check on fills

Table 8: Production bot specification. Identical logic on MES and MNQ; params are those of VWAP fade variant 1.

7.2 Launch sequence

Run the strategy in NT8 Market Replay and then on a single sim account for two weeks, verifying that VWAP-target limit orders actually fill on touch-through. Buy one Topstep 50K with the daily loss limit selected at checkout, run the bot supervised, and treat the first evaluation as a live validation of the degradation haircut: if 20 traded days produce an average day better than the stress scenario's, the model's confidence intervals apply as published. On passing, pay activation, trade the funded account to the first payout at the same size, withdraw at the cap, and only then consider scaling - a second Topstep account or an Apex 50K running the same logic (supervised; no Apex copy-trading). Re-verify firm rules monthly; every parameter in Table 4 changed at least once during 2025-2026.

8. Limitations and Risks

Honest limits of this research. **Backtest realism:** hourly bars cannot resolve every intrabar sequence, so even conservative fill models (stop-first, trade-through targets, 0.5-tick slippage) can be beaten by live fills; the base and stress haircuts absorb this, but they are assumptions, not guarantees, and a two-week Market Replay shakedown is the cheapest insurance before real evaluation fees are at stake. **Regime coverage:** the hourly history spans June 2024 to November 2026 - a high-volatility, trending period; a multi-year range-bound regime would change the VWAP fade's profile, which is exactly why position sizes stay small and the daily stop stays hard. **Bootstrap assumptions:** block bootstrap preserves short-range persistence but cannot foresee rule changes or a firm simply refusing a payout, and payout caps mean large funded profits take several weekly cycles to fully withdraw. **Rule volatility:** every firm revised rules and prices repeatedly during 2025-2026 (Apex's March 2026 one-time restructure and MFFU's August 2026 plan changes are two recent examples); re-confirm every fee and rule at purchase. **Compliance:** Apex's semi-auto-only policy makes unattended operation a payout risk; Topstep's post-payout floor reset concentrates bust risk after you withdraw. Finally, this document is a quantitative research exercise built on simulated evaluations and free historical data - it is not financial advice, futures trading carries substantial risk of loss, and past or simulated performance never guarantees future results.